

WEB FUNCTIONAL TESTING & SELENIUM IDE AUTOMATION

TNC Store E-commerce Website

Major: Information Technology

Student's name:

1. Pham Duc Duy ID: 2280618268 Class: 22DTHQB3

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- Test Case Summary:

TC ID	TC Description	Author	Mark
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- **Test Case Details:** (Students may use either Vietnamese or English)

TC ID	TC Description	Steps	Expected Result	Test Result
TC-01	Verify Successful Login with Valid Credentials	1. Navigate to: https://www.tncstore.vn/ 2. Click on the Account icon/menu. 3. Enter login credentials: 4. Email: [REDACTED TEST ACCOUNT] 5. Password: [REDACTED] 6. Press Enter key or click the Login button. 7. Verify successful login status.	The Đăng Xuất button is displayed (indicating successful login).	Pass
TC-02	Verify Successful Logout from Account	1. Navigate to: https://www.tncstore.vn/ 2. Click on the Account icon/menu. 3. Check if the Đăng xuất (Logout) text is visible. 4. If visible, click on Đăng xuất. 5. Verify the logout status.	The Tài khoản (Account) link/button is displayed again, confirming the user is successfully logged out.	Pass
TC-03	Verify Cart Quantity Updates When Adding Products	1. Navigate to: https://www.tncstore.vn/ 2. Select menu PC Gaming. 3. Add all available products to the shopping cart. 5. Verify the cart quantity displayed on the homepage.	The cart quantity displayed matches the number of products added.	Pass
TC-04	Verify Empty Cart State After Removing All Items	1. Navigate to: https://www.tncstore.vn/ 2. Check the current number of products in the cart on the homepage. (*) Advanced Scenario (Dynamic Handling): If the cart contains products (Quantity > 0): Click on the Shopping cart icon and navigate to the cart details page. Remove all products from the cart one by one by clicking the "X" (remove) button until no products are left. Verify the cart status message	- The cart is successfully cleared of all items. - The message "Ôi! Bạn chưa có sản phẩm vào giỏ hàng!" is displayed correctly on the screen.	Pass

		If the cart is already empty (Quantity = 0): End the test successfully without navigating to the cart.		
TC-05	Verify Product Sorting by Price (Low to High)	1. Navigate to: https://www.tncstore.vn/ 2. Access a specific product category page from the main menu. 3. Open the Sort dropdown menu and select the "Giá: Tăng dần" (Price: Low to High) option. 4. Continuously click the "Xem thêm" (Load More) button until all available products in the chosen category are fully loaded on the screen. 5. Scan through the entire displayed product list. 6. Sequentially extract and compare the price of each product with the price of the immediately following product.	Product prices are sorted correctly from lowest to highest across all displayed products.	Pass
TC-06	Verify Product Filtering Functionality by Brand	1. Navigate to the homepage: https://www.tncstore.vn/ 2. Click on a product category menu (e.g., Laptops/Monitors). 3. Locate the product filter section on the left sidebar or top. 4. Select a specific attribute filter option (e.g., Check the box for Brand: Dell). 5. Wait for the product grid to refresh and apply the filter. 6. Count the total number of products displayed on the page. 7. Iterate through all displayed products and perform the following checks: • Extract the product name. • Verify that every product name contains the selected	• The product list must update successfully and only display items that match the selected attribute. • 100% of the displayed products must belong to the selected brand (Dell). • Products from other unselected brands must be completely hidden from the result list.	Pass

		attribute keyword (e.g., "Dell").		
TC-07	Verify Consistency of Product Count in Category	1. Navigate to the homepage: https://www.tncstore.vn/ 2. Click on a product category from the navigation menu to open its listing page. 3. Locate the label that displays the total number of products for this category (e.g., "Có 150 sản phẩm"). 4. Extract and record the Expected Total number from this label. 5. Scroll down to the bottom of the product grid. 6. Iterate through the following steps until all products are loaded: • Check if the "Xem thêm" (Load More) button is visible. • If visible, click the button to load the next batch of products. • Wait for the new products to render. 7. Once the "Xem thêm" button is no longer available, count the Actual Total number of product item elements displayed on the page. 8. Verify that the Actual Total count matches the Expected Total number extracted in step 4	• The "Xem thêm" functionality must load subsequent products without any UI breaks or duplicate items. • The system must display the exact number of products as advertised. • The total count of actual product elements rendered on the page must strictly equal the expected total number shown on the category's header label.	Pass
TC-08	Verify Alert for Exceeding Maximum Purchase Limit	1. Navigate to the homepage: https://www.tncstore.vn/ 2. Hover the mouse over the Cart icon in the header (#js-header-cart). 3. Click the Xem giỏ hàng (View Cart) link from the dropdown to navigate to the Cart page.	• The system must actively monitor the quantity input and prevent orders exceeding the limit.	Pass

		4. Locate the quantity input field for the product (id=js-buy-quantity). 5. Type an excessive quantity value that exceeds the maximum allowed purchase limit (e.g., 99). 6. Verify the system's immediate response and handle the browser alert.	• A browser alert (JavaScript prompt) must immediately appear with the exact warning message: "Bạn được mua tối đa 10 sản phẩm này". • The product quantity must not be updated to the invalid excessive number.	
TC-09	Verify Accurate Calculation of Cart Total Amount	1. Navigate to: https://www.tncstore.vn/ 2. Select the Laptop Gaming category menu. 3. For the all products in the list, repeat the following steps: • Record the displayed price of the product. • Click on the product to open its detail page. • Click the Mua ngay (Buy Now) button. • Navigate back to the Laptop Gaming category page. 4. Calculate the expected total amount by summing up the recorded prices of the 3 products. 5. Navigate to the Cart page (/cart). 6. Verify the total price displayed in the cart against the calculated expected total.	• The cart successfully stores all selected products. • The actual Total Amount displayed in the Cart matches exactly with the calculated sum of the individual product prices.	Pass
TC-10	Verify Change Password Failure with Incorrect Current Password	1. Navigate to: https://www.tncstore.vn/taikhoan 2. Click on the Thay đổi mật khẩu (Change Password) tab. 3. Fill in the change password form:	• The system must reject the password change request and ideally show an error message. • The password must remain	Pass

		• Current password: Enter an incorrect old password. • New password: Enter a valid new password. • Confirm new password: Enter the new password again. 4. Click the Submit / Update button. 5. Log out of the current account. 6. Navigate back to the Login page. 7. Attempt to log in using the account's email and the New Password entered in step 3. 8. Verify the login result and the system's behavior.	unchanged in the database. • Attempting to log in with the new password in Step 7 must fail (The user should not be able to access the account).	
TC-11	Verify Search Results Relevance by Keyword	1. Navigate to the homepage: https://www.tncstore.vn/ 2. Locate the global search bar (id=js-global-seach). 3. Input a specific keyword (e.g., "ASUS ROG"). 4. Click the Search button and wait for the search results page to fully load. 5. Count the total number of products displayed in the search result grid (#js-product-list). 6. Iterate through all displayed products and perform the following checks: • Extract the product name/title of the current item. • Convert the text to lowercase for case-insensitive comparison. • Verify that the product name contains the exact searched keyword(s) (e.g., "asus rog"). 7. Log an error for any product that fails the keyword match check.	• The system must successfully return a list of products based on the inputted keyword. • 100% of the products displayed in the result list must contain the searched keyword(s) in their titles, ensuring the search engine's accuracy. • No unrelated products should be mixed into the search results.	Pass

TC-12	Verify Dynamic Search Suggestions by Keyword	1. Navigate to the homepage: https://www.tncstore.vn/ 2. Locate the global search input field. 3. Type a specific keyword (e.g., "ASUS ROG") into the search bar without submitting/pressing Enter. 4. Wait for the search suggestion dropdown box to dynamically appear on the screen. 5. Count the total number of suggested items loaded within the dropdown. 6. Iterate through every suggested product in the list and perform the following checks: • Extract the name/title of the suggested product. • Convert the text for case-insensitive comparison. • Verify that the suggested product name contains the inputted keyword(s).	• The system must dynamically display a dropdown list of product suggestions shortly after the user types the keyword. • 100% of the suggested items displayed in the dropdown must be highly relevant and contain the searched keyword(s). • The suggestions must match the keyword accurately regardless of uppercase or lowercase letters.	Pass
TC-13	Verify Data Persistence After Updating Account Details	1. Navigate to the Account Information page. 2. Enter new valid data into the Name and Phone Number fields. 3. Click the Cập nhật / Lưu (Update/Save) button. 4. Log out of the current session. 5. Log in again using the account's current credentials. 6. Navigate back to the Account Information page. 7. Verify the values currently displayed in the Name and Phone Number fields.	• The system saves the changes successfully without any errors. • Upon re-login, the input fields strictly retain the newly updated information, confirming that the backend database was accurately updated and data is persistent.	Pass

TC-14	Verify Cart Quantity Capping with Excessive Input	1. Navigate to the Cart page: https://www.tncstore.vn/cart 2. Wait for the cart product list to fully load. 3. Count the total number of distinct products currently in the cart. 4. For each product in the cart, repeat the following steps: • Click on the quantity input field. • Type an excessively huge number (e.g., 9999). • Press the Enter key to submit the new quantity. • Wait for the system to process the update and refresh the cart. • Retrieve the newly updated quantity from the input field. 5. Verify the final quantity: • Check that the actual quantity is <i>different</i> from the inputted huge value. • Check that the actual quantity is strictly <i>less than</i> the inputted huge value (meaning the system successfully capped it at the maximum available stock limit).	• The system must not accept the excessively large quantity input (9999). • The system must automatically restrict and adjust the quantity down to the maximum allowed limit (e.g., maximum available in stock). • The total price of the cart must be calculated based on this capped quantity, not the invalid user input.	Pass
TC-15	Verify Internal Chat Widget Initialization and Expansion	1. Navigate to the TNC Store homepage: https://www.tncstore.vn/ 2. Wait for the page to fully load and locate the internal chat widget icon (usually at the bottom right corner). 3. Click on the chat widget icon. 4. Verify the chat window interface. 5. Click the minimize/close ('X') button on the chat window.	• The chat window expands successfully upon clicking. • Elements such as the welcome message and input text area are visible. • The window collapses back to an icon when closed.	Pass

TC-16	Verify Mandatory Field Validations on Product Review Form (Negative Test)	1. Navigate to a specific product detail page. 2. Click on the button to write a review (opening the review modal). 3. Leave all mandatory fields empty (Rating, Content, Name, Phone). 4. Click the "Gửi đánh giá" (Submit Review) button. 5. Observe the validation messages displayed under each field.	The system prevents the submission and displays specific inline error messages: - "Nội dung quá ngắn". - "Tên quá ngắn". - "Bạn chưa nhập SĐT".	Pass
TC-17	Verify Successful Product Review Submission (Positive Test)	1. Navigate to a specific product detail page. 2. Click on the button to write a review. 3. Select a star rating (e.g., 5 stars). 4. Enter valid text into the review content field. 5. Enter a valid name and phone number. 6. Click the "Gửi đánh giá" (Submit Review) button. 7. Observe the system's response.	The system successfully accepts the review. A success confirmation message or popup is displayed, and the review form closes.	Pass
TC-18	Verify Dynamic Product Image Gallery Functionality	1. Navigate to a product detail page. 2. Observe the default main product image. 3. Click on a different thumbnail in the image gallery. 4. Verify that the main image updates to match the clicked thumbnail. 5. Repeat the process for all available thumbnails.	The system successfully updates the main display image corresponding to the clicked thumbnail without reloading the page.	Pass
TC-19	Verify "Build PC" Redirection with Pre-selected Component	1. Navigate to a specific component's product detail page (e.g., VGA). 2. Record the name of the product. 3. Click the "Thêm vào cấu hình" (Add to Build PC) button. 4. Verify redirection to the Build PC page.	The system successfully redirects the user to the Build PC page and accurately pre-selects the chosen component without any data	Pass

		- Record the name of the component populated in the Build PC list. - Verify that the product name from the detail page matches the pre-selected component name in the builder.	loss or mismatch.	
TC-20	Verify Dynamic Price Deduction (PC Builder)	- Navigate to the PC Builder/Configuration page with pre-selected items. - Record the initial Total Price and the specific Price of one selected item. - Click the remove/delete icon for that specific item. - Accept the browser confirmation popup to proceed with the removal. - Record the updated Total Price after the UI refreshes. - Verify that the new total is mathematically correct (New Total = Initial Total - Removed Item Price).	The system successfully removes the item and dynamically deducts its exact price from the total configuration cost without any calculation errors.	Pass
TC-21	Validate Min-Max Price Filter	- Navigate to a specific product category page: https://www.tncstore.vn/gaming-laptop.html - Select a specific price range option from the filter sidebar. - Wait for the product list to fully reload and update according to the applied filter. - Scan through the entire displayed product list. - Sequentially extract the price of each product and verify that it falls within the applied minimum and maximum price boundaries.	All displayed product prices are strictly within the selected minimum and maximum price range.	Pass
TC-22	Verify Cart Item Accumulation and Total Price Calculation	- Navigate to a specific product category page: https://www.tncstore.vn/gaming-laptop.html - Select a specific product and click the "Buy Now" (Add to Cart) button twice in succession.	The cart must display only one row for the product, avoiding duplicate rows for the same item.	Pass

		3. Navigate to the Shopping Cart page (/cart). 4. Verify the number of displayed item rows in the cart against the number of unique products added. 5. Check the accumulated quantity (sl) for the added product. 6. Sequentially calculate the expected total price by multiplying each item's price with its quantity. 7. Compare the calculated total price with the website's displayed total price.	• The product's quantity input field must show a value of 2. • The manually calculated total price must strictly match the website's displayed Total Price.	
TC-23	Verify Total Price Deduction upon Item Removal	1. Navigate to the Shopping Cart page (/cart) containing at least one product. 2. Extract the initial Total Price (old_tot) and the specific details (Price and Quantity) of the first item in the cart. 3. Click the "Remove" (X) button for that specific first item and confirm the deletion prompt (if any). 4. Wait for the cart to reload and update the item count. 5. Extract the newly updated Total Price from the website (new_tot). 6. Manually calculate the expected total: Subtract the removed item's total value (Price * Quantity) from the initial Total Price. 7. Compare the manually calculated expected total with the website's updated new_tot.	• The selected item is successfully removed from the cart. • The displayed Total Price is dynamically and accurately reduced. • The manually calculated total (expected_tot) must perfectly match the newly displayed Total Price (new_tot).	Pass
TC-24	Discount Logic Validation	1. Navigate to a product details page that has an active promotional discount. 2. Extract the crossed-out Original Price (oldPrice) and clean any non-numeric characters.	• The Original Price, Promotional Price, and Discount Percentage are successfully extracted and	Pass

		3. Extract the current Promotional/Discounted Price (newPrice) and clean any non-numeric characters. 4. Extract the Discount Percentage label (webDisc) and clean any non-numeric characters (removing '-' and '%'). 5. Manually calculate the expected discount percentage using the ceiling rounding formula: $\text{Math.ceil}(((\text{oldPrice} - \text{newPrice}) / \text{oldPrice}) * 100)$ to get the calcDisc. 6. Compare the manually calculated percentage (calcDisc) with the website's displayed percentage (webDisc).	converted into valid numeric formats. • The manually calculated discount percentage (calcDisc) must perfectly match the website's displayed Discount Percentage (webDisc).	
TC-25	Verify Build PC Total Price After Editing	1. Navigate to the PC Builder page (/buildpc). 2. Sequentially edit multiple components by selecting a different item from the popup. 3. For each swap, manually calculate the expected total: $(\text{Old Total} - \text{Old Item Price}) + \text{New Item Price}$ 4. Compare the calculated expected total with the newly displayed Total Price on the website.	• The displayed Total Price updates dynamically and accurately upon changing any component. • The calculated expected total must strictly match the website's updated Total Price.	Pass
TC-26	Verify Discount For Complete PC	1. Navigate to the PC Builder page (/buildpc). 2. Scan and auto-fill any missing mandatory components to trigger the bundle discount. 3. Temporarily remove one component to test the discount revocation, then re-add it to restore the bundle. 4. Sequentially edit multiple components from the list.	• The 5% discount is dynamically revoked when any mandatory component is missing. • The applied discount amount is perfectly accurate (exactly 5% of the Subtotal).	Pass

		- For each update, extract the Subtotal, Discount, and Total Price. - Verify that the applied Discount strictly equals 5% of the Subtotal. - Verify the mathematical equation: Subtotal - Discount = Total Price (allowing a ≤ 50 VND rounding tolerance).	The final Total Price is mathematically correct without any calculation errors across multiple component changes.	
TC-27	Verify Search Results using Inline Search Bar Category Filter	- Navigate to the website's homepage. - Click on the Category Dropdown embedded directly <i>inside</i> the global search bar (#js-search-title). - Select a specific category from this dropdown (e.g., "Laptop - Máy Tính Xách Tay"). - Enter a search keyword (e.g., "ASUS") into the search input field and submit the query. - Continuously click the "Load More" (Xem thêm) button until all matching products are fully rendered on the page. - Iterate through the entire product list and extract each product's name. - Validate that every extracted name contains both the selected category identifier ("Laptop") and the keyword ("ASUS"), using case-insensitive normalization. - Log any mismatched products and output the total bug count at the end.	- The inline search bar accurately passes both the category parameter and the search keyword to the results page. - Not 100% of the displayed products strictly match the combined criteria (containing both "Laptop" and "ASUS"). - The script completely scans all items and provides a full bug report without prematurely aborting.	Pass
TC-28	Verify "Reset Configuration" Functionality in PC Builder	- Navigate to the homepage (/). - Click on a pre-built PC template from the carousel to auto-populate the PC builder list.	The pre-built PC template successfully loads multiple components into the list.	Pass

		3. Extract and store the total number of populated components (count_before) using the .product-name selector. 4. Click the "Reset Configuration" (Làm mới cấu hình) button. 5. Handle the confirmation prompt by clicking the confirmation button. 6. Extract the component count again (count_after). 7. Verify that the final count is strictly equal to 0.	• The confirmation prompt appears upon clicking the reset button. • Upon confirmation, the system completely clears all selected components from the list. • No product names remain visible in the layout, reflecting a completely empty state (count_after = 0).	
TC-29	Verify Shopping Cart Data is Retained Post-Authentication	1. Navigate to the homepage and verify the authentication state. If currently logged in, perform a logout to ensure a clean "Guest" session. 2. Browse to a product category, select the first available product, and click "Buy Now" to add it to the cart. 3. Extract and securely store the name of the product currently in the guest cart (guestItem). 4. Navigate to the authentication form, input valid user credentials (email and password), and submit to log in. 5. Navigate directly to the shopping cart page (/cart). 6. Extract the name of the product currently displayed in the authenticated user's cart (loggedItem).	• The system successfully adds the selected product to the temporary guest session cart. • User authentication is processed successfully without errors. • The system successfully merges the guest cart data with the authenticated user's account. • The exact product added during the guest session is persistently displayed in the cart after	Pass

		7. Sanitize both strings (convert to lowercase, remove special characters) and verify that the loggedItem string contains the core identifier (first 15 characters) of the guestItem.	logging in (String validation evaluates to true).	
TC - 30	Data Integrity Validation (Product Name vs. Detailed Specs)	1. Navigate to the product category page and retrieve all product links to iterate through. 2. For each product, navigate to the product details page and extract the Product Name. 3. Verify the existence of the detailed specifications expand button (e.g., "Xem thêm"). If it does not exist, log a "SKIP" status and proceed to the next product. 4. If the button exists, click it, wait for the table to render, and extract the full Detailed Specifications text. 5. Clean and normalize the data: Extract core specifications (e.g., RAM, Storage capacity, GPU series) from the Product Name using Regex. Strip out measurement units (GB, TB, Hz, etc.) and special characters from both the extracted core specs and the Detailed Specifications text. 6. Verify that every normalized core specification extracted from the Product Name is present within the normalized Detailed Specifications text. 7. Log the final verification result (Pass for consistent data,	1. The script successfully iterates through the entire list of products without timing out or crashing due to missing elements. 2. Core specifications are successfully extracted, stripped of formatting/units, and converted into comparable strings to prevent false negatives. 3. The logical comparison accurately identifies and logs "Pass" for consistent data, "Fail" for mismatched data (Data Inconsistency), and "Skip" for products missing the specification table.	

		Fail for mismatched data) for the current product.		
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